

Eigen Values and Vectors

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Problem Statement

What can we conclude if a matrix **A** is symmetric?

Proof

Proof :

Case 1 : $\mathbf{A}$ is real and symmetric

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be a real symmetric matrix, i.e.,

$$\mathbf{A} = \mathbf{A}^\top \quad (0.1)$$

Suppose λ is an eigenvalue of $\mathbf{A}$ with eigenvector $\mathbf{v} \neq \mathbf{0}$:

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v} \quad (0.2)$$

Taking the conjugate transpose of $\mathbf{v}$ and multiplying on the left:

$$\mathbf{v}^* \mathbf{A} \mathbf{v} = \lambda \mathbf{v}^* \mathbf{v} \quad (0.3)$$

Since $\mathbf{A}$ is real symmetric, $\mathbf{A} = \mathbf{A}^* = \mathbf{A}^\top$, therefore:

$$\mathbf{v}^* \mathbf{A} \mathbf{v} = (\mathbf{A} \mathbf{v})^* \mathbf{v} = (\lambda \mathbf{v})^* \mathbf{v} = \bar{\lambda} \mathbf{v}^* \mathbf{v} \quad (0.4)$$

Equating both expressions for $\mathbf{v}^* \mathbf{A} \mathbf{v}$:

$$\lambda \mathbf{v}^* \mathbf{v} = \bar{\lambda} \mathbf{v}^* \mathbf{v} \quad (0.5)$$

$$(\lambda - \bar{\lambda}) \mathbf{v}^* \mathbf{v} = \mathbf{0} \quad (0.6)$$

Proof

Since $\mathbf{v} \neq \mathbf{0}$, $\mathbf{v}^* \mathbf{v} > 0$. Hence:

$$\lambda = \bar{\lambda} \quad (0.7)$$

Therefore, every eigenvalue λ of $\mathbf{A}$ is real.

Eigenvectors corresponding to distinct eigenvalues are orthogonal

Let

$$\mathbf{A}\mathbf{u} = \alpha\mathbf{u} \quad (0.8)$$

$$\mathbf{A}\mathbf{w} = \beta\mathbf{w} \quad (0.9)$$

Where $\alpha \neq \beta$ are distinct eigenvalues. Multiplying the first equation on the left by $\mathbf{w}^\top$:

$$\mathbf{w}^\top \mathbf{A}\mathbf{u} = \alpha \mathbf{w}^\top \mathbf{u} \quad (0.10)$$

Proof

Since $\mathbf{A}$ is symmetric, $\mathbf{A} = \mathbf{A}^\top$. Hence:

$$\mathbf{w}^\top \mathbf{A} \mathbf{u} = (\mathbf{A} \mathbf{w})^\top \mathbf{u} = (\beta \mathbf{w})^\top \mathbf{u} = \beta \mathbf{w}^\top \mathbf{u} \quad (0.11)$$

Equating both forms:

$$\alpha \mathbf{w}^\top \mathbf{u} = \beta \mathbf{w}^\top \mathbf{u} \quad (0.12)$$

Simplifying:

$$(\alpha - \beta) \mathbf{w}^\top \mathbf{u} = 0 \quad (0.13)$$

Since $\alpha \neq \beta$, it follows that:

$$\mathbf{w}^\top \mathbf{u} = 0 \quad (0.14)$$

Hence, eigenvectors corresponding to distinct eigenvalues are orthogonal.

Proving that Eigen Vector is real

Suppose $\lambda \in \mathbb{R}$ is an eigenvalue of $\mathbf{A}$ with a (possibly complex) eigenvector $\mathbf{v} \in \mathbb{C}^n$, satisfying

$$\mathbf{A} \mathbf{v} = \lambda \mathbf{v} \quad (0.15)$$

Proof

Considering the conjugate eigenvector

Since $\mathbf{A}$ is real, taking the complex conjugate of both sides gives

$$\mathbf{A}\bar{\mathbf{v}} = \bar{\lambda}\bar{\mathbf{v}} \quad (0.16)$$

But for a real symmetric matrix, $\lambda = \bar{\lambda}$ (as proven earlier). Hence,

$$\mathbf{A}\bar{\mathbf{v}} = \lambda\bar{\mathbf{v}} \quad (0.17)$$

Thus, $\bar{\mathbf{v}}$ is also an eigenvector corresponding to the same eigenvalue λ .

Constructing real vectors from $\mathbf{v}$ and $\bar{\mathbf{v}}$

Define

$$\mathbf{u} = \mathbf{v} + \bar{\mathbf{v}} \quad (0.18)$$

$$\mathbf{w} = i(\mathbf{v} - \bar{\mathbf{v}}) \quad (0.19)$$

Both $\mathbf{u}$ and $\mathbf{w}$ are **real vectors**, because $\mathbf{u}$ is the sum of a vector and its conjugate, and $\mathbf{w}$ is i times their difference.

Proof

Applying $\mathbf{A}$ to $\mathbf{u}$ and $\mathbf{w}$

Since $\mathbf{A}$ is real and linear,

$$\mathbf{A}\mathbf{u} = \mathbf{A}(\mathbf{v} + \bar{\mathbf{v}}) = \mathbf{A}\mathbf{v} + \mathbf{A}\bar{\mathbf{v}} = \lambda\mathbf{v} + \lambda\bar{\mathbf{v}} = \lambda(\mathbf{v} + \bar{\mathbf{v}}) = \lambda\mathbf{u} \quad (0.20)$$

Similarly,

$$\mathbf{A}\mathbf{w} = \mathbf{A}(i(\mathbf{v} - \bar{\mathbf{v}})) = i(\mathbf{A}\mathbf{v} - \mathbf{A}\bar{\mathbf{v}}) = i(\lambda\mathbf{v} - \lambda\bar{\mathbf{v}}) = \lambda i(\mathbf{v} - \bar{\mathbf{v}}) = \lambda\mathbf{w} \quad (0.21)$$

Hence we get

$$\mathbf{A}\mathbf{u} = \lambda\mathbf{u} \quad (0.22)$$

$$\mathbf{A}\mathbf{w} = \lambda\mathbf{w} \quad (0.23)$$

Real eigenvectors

If $\mathbf{u} \neq \mathbf{0}$, then $\mathbf{u}$ is a real eigenvector corresponding to eigenvalue λ . If $\mathbf{w} \neq \mathbf{0}$, then $\mathbf{w}$ is a real eigenvector corresponding to eigenvalue λ .

Proof

Conclusion : For any real symmetric matrix $\mathbf{A}$,

- 1) All eigenvalues of $\mathbf{A}$ are **real**.
- 2) Eigenvectors corresponding to distinct eigenvalues are **orthogonal**.
- 3) For every eigenvalue λ of a real symmetric matrix $\mathbf{A}$, there exists a **real eigenvector**. Hence, the eigenvectors of real symmetric matrices are real.

Case 2 : $\mathbf{A}$ is complex and symmetric

Let $\mathbf{A} \in \mathbb{C}^{n \times n}$ be a complex symmetric matrix satisfying

$$\mathbf{A} = \mathbf{A}^T \quad (0.24)$$

Note that $\mathbf{A}$ is not necessarily Hermitian since in general $\mathbf{A} \neq \mathbf{A}^*$.

$$\mathbf{A} \neq \mathbf{A}^* \quad (0.25)$$

Eigenvalues of $\mathbf{A}$ may be complex

Let $\lambda \in \mathbb{C}$ be an eigenvalue of $\mathbf{A}$ with eigenvector $\mathbf{v} \neq \mathbf{0}$ such that

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v} \quad (0.26)$$

Proof

Define scalar z as :

$$z = \mathbf{v}^* \mathbf{A} \mathbf{v} \quad (0.27)$$

Taking complex conjugate transpose on both sides ,

$$z^* = \mathbf{v}^* \mathbf{A}^* \mathbf{v} \quad (0.28)$$

If $\mathbf{A} = \mathbf{A}^*$, then

$$z = z^* \quad (0.29)$$

$$z = \bar{z} \text{ as } z \text{ is a scalar} \quad (0.30)$$

Therefore z is a real scalar value

For Hermitian matrices ($\mathbf{A} = \mathbf{A}^*$), the quantity $\mathbf{v}^* \mathbf{A} \mathbf{v}$ is always real.

However, for complex symmetric matrices, $\mathbf{A} \neq \mathbf{A}^*$, so

Proof

$$\mathbf{v}^* \mathbf{A} \mathbf{v} \text{ may not be real.} \quad (0.31)$$

The Rayleigh quotient is

$$\frac{\mathbf{v}^* \mathbf{A} \mathbf{v}}{\mathbf{v}^* \mathbf{v}} = \lambda \quad (0.32)$$

Since $\mathbf{v}^* \mathbf{A} \mathbf{v}$ may be complex, λ may also be complex.

Eigenvectors corresponding to distinct eigenvalues need not be orthogonal

For complex symmetric matrices, we have only $\mathbf{A} = \mathbf{A}^\top$, so the Hermitian symmetry property does not hold.

Thus, for two eigenpairs $(\lambda_1, \mathbf{v}_1)$ and $(\lambda_2, \mathbf{v}_2)$ with $\lambda_1 \neq \lambda_2$, we **cannot guarantee**

$$\mathbf{v}_1^* \mathbf{v}_2 = 0 \quad (0.33)$$

Proof

Let

$$\mathbf{A}\mathbf{v}_1 = \alpha\mathbf{v}_1 \quad (0.34)$$

$$\mathbf{A}\mathbf{v}_2 = \beta\mathbf{v}_2 \quad (0.35)$$

Where $\alpha \neq \beta$ are distinct eigenvalues. Multiplying the first equation on the left by $\mathbf{v}_2^*$:

$$\mathbf{v}_2^* \mathbf{A} \mathbf{v}_1 = \alpha \mathbf{v}_2^* \mathbf{v}_1 \quad (0.36)$$

As $\mathbf{A} \neq \mathbf{A}^*$ and also $\mathbf{A} \neq \bar{\mathbf{A}}$ we cannot proceed like what we had done earlier for **case 1**.

Hence, eigenvectors corresponding to distinct eigenvalues **need not be orthogonal and are not guaranteed to be real**.

Conclusion : For a complex symmetric matrix $\mathbf{A}$:

- 1) The eigenvalues **can be complex**.
- 2) The eigenvectors corresponding to distinct eigenvalues are **not necessarily orthogonal and need not be real**.

But if $\mathbf{A} = \mathbf{A}^*$ then we get the same result as **case 1**

Examples

Examples to Support the Proof

Example 1 : Real and Symmetric Matrix

(a) Let

$$\mathbf{A} = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix} \quad (0.37)$$

Clearly, $\mathbf{A} = \mathbf{A}^T$, hence $\mathbf{A}$ is real symmetric.

Finding Eigenvalues

The characteristic equation is:

$$|\mathbf{A} - \lambda \mathbf{I}| = 0 \quad (0.38)$$

Substituting $\mathbf{A}$,

$$\begin{vmatrix} 2 - \lambda & -1 & 0 \\ -1 & 2 - \lambda & -1 \\ 0 & -1 & 2 - \lambda \end{vmatrix} = 0 \quad (0.39)$$

Examples

$$\begin{vmatrix} 2-\lambda & -1 & 0 \\ -1 & 2-\lambda & -1 \\ 0 & -1 & 2-\lambda \end{vmatrix} \xleftrightarrow{R_2 \rightarrow R_2 + \frac{1}{2-\lambda} R_1} \begin{vmatrix} 2-\lambda & -1 & 0 \\ 0 & \frac{(2-\lambda)^2-1}{2-\lambda} & -1 \\ 0 & -1 & 2-\lambda \end{vmatrix} \xleftrightarrow{R_3 \rightarrow R_3 + \frac{(2-\lambda)}{(2-\lambda)^2-1} R_2} \quad (0.40)$$

$$(2-\lambda) \left(\frac{(2-\lambda)^2-1}{2-\lambda} \right) \left(\frac{(2-\lambda)^3-4(2-\lambda)}{(2-\lambda)^2-1} \right) = 0 \quad (0.41)$$

Simplifying,

$$(2-\lambda)((2-\lambda)^2-2) = 0 \quad (0.42)$$

Hence, the eigenvalues are:

$$\lambda_1 = 2, \lambda_2 = 2 + \sqrt{2}, \lambda_3 = 2 - \sqrt{2} \quad (0.43)$$

All eigenvalues are real, confirming positive definiteness.

Finding Eigenvectors

Examples

For $\lambda_1 = 2$:

$$(\mathbf{A} - 2\mathbf{I})\mathbf{v} = 0 \quad (0.44)$$

$$\begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad (0.45)$$

$$\left(\begin{array}{ccc|c} 0 & -1 & 0 & 0 \\ -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 \end{array} \right) \xleftrightarrow{R_2 \leftrightarrow R_1} \left(\begin{array}{ccc|c} -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{array} \right) \quad (0.46)$$

$$\xleftrightarrow{R_3 \rightarrow R_3 - R_2} \left(\begin{array}{ccc|c} -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \quad (0.47)$$

Hence, $x = -z$, $y = 0 \Rightarrow \mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$.

Examples

For $\lambda_2 = 2 + \sqrt{2}$:

$$(\mathbf{A} - (2 + \sqrt{2})\mathbf{I})\mathbf{v} = 0 \quad (0.48)$$

$$\begin{pmatrix} -\sqrt{2} & -1 & 0 \\ -1 & -\sqrt{2} & -1 \\ 0 & -1 & -\sqrt{2} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad (0.49)$$

$$\left(\begin{array}{ccc|c} -\sqrt{2} & -1 & 0 & 0 \\ -1 & -\sqrt{2} & -1 & 0 \\ 0 & -1 & -\sqrt{2} & 0 \end{array} \right) \xleftrightarrow{R_2 \rightarrow R_2 - \frac{1}{\sqrt{2}}R_1} \left(\begin{array}{ccc|c} -\sqrt{2} & -1 & 0 & 0 \\ 0 & -\frac{1}{\sqrt{2}} & -1 & 0 \\ 0 & -1 & -\sqrt{2} & 0 \end{array} \right) \quad (0.50)$$

$$\xleftrightarrow{R_3 \rightarrow R_3 - \sqrt{2}R_2} \left(\begin{array}{ccc|c} -\sqrt{2} & -1 & 0 & 0 \\ 0 & -\frac{1}{\sqrt{2}} & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \quad (0.51)$$

Examples

Simplifying gives proportional relation $y = -\sqrt{2}x$, $z = x$. Hence

$$\mathbf{v}_2 = \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix}.$$

Examples

For $\lambda_3 = 2 - \sqrt{2}$: Similarly, solving gives $\mathbf{v}_3 = \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix}$.

Thus, eigenvectors are **real and mutually orthogonal**.

Example 2 : Complex Symmetric Matrix

(a) Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1+i \\ 1+i & 1 \end{pmatrix} \quad (0.52)$$

Here, $\mathbf{A} = \mathbf{A}^\top$ but $\mathbf{A} \neq \mathbf{A}^*$, so $\mathbf{A}$ is complex symmetric but not Hermitian.

Finding Eigenvalues

$$|\mathbf{A} - \lambda \mathbf{I}| = 0 \quad (0.53)$$

$$\begin{vmatrix} 1 - \lambda & 1 + i \\ 1 + i & 1 - \lambda \end{vmatrix} = 0 \quad (0.54)$$

Examples

$$\left| \begin{array}{cc} 1 - \lambda & 1 + i \\ 1 + i & 1 - \lambda \end{array} \right| \xleftrightarrow{R_2 \rightarrow R_2 - \frac{1+i}{1-\lambda} R_1} \left| \begin{array}{cc} 1 - \lambda & 1 + i \\ 0 & (1 - \lambda) - \frac{(1+i)^2}{1-\lambda} \end{array} \right| \quad (0.55)$$

$$(1 - \lambda)^2 - (1 + i)^2 = 0 \quad (0.56)$$

$$(1 - \lambda)^2 - 2i = 0 \quad (0.57)$$

$$1 - \lambda = \pm \sqrt{2i} \quad \Rightarrow \quad \lambda = 1 \mp \sqrt{2i} \quad (0.58)$$

Hence eigenvalues are:

$$\lambda_1 = 1 - \sqrt{2i}, \quad \lambda_2 = 1 + \sqrt{2i} \quad (0.59)$$

Examples

Finding Eigenvectors

For $\lambda_1 = 1 - \sqrt{2}i$:

$$(\mathbf{A} - \lambda_1 \mathbf{I})\mathbf{v} = 0 \quad (0.60)$$

$$\begin{pmatrix} \sqrt{2}i & 1+i \\ 1+i & \sqrt{2}i \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (0.61)$$

$$\left(\begin{array}{cc|c} \sqrt{2}i & 1+i & 0 \\ 1+i & \sqrt{2}i & 0 \end{array} \right) \xleftrightarrow{R_2 \rightarrow R_2 - \frac{1+i}{\sqrt{2}i} R_1} \left(\begin{array}{cc|c} \sqrt{2}i & 1+i & 0 \\ 0 & 0 & 0 \end{array} \right) \quad (0.62)$$

Hence $y = -\frac{\sqrt{2}i}{1+i}x$ and $\mathbf{v}_1 = \begin{pmatrix} 1 \\ -\frac{\sqrt{2}i}{1+i} \end{pmatrix}$.

Examples

For $\lambda_2 = 1 + \sqrt{2}i$:

$$(\mathbf{A} - \lambda_2 \mathbf{I})\mathbf{v} = 0 \quad (0.63)$$

$$\begin{pmatrix} -\sqrt{2}i & 1+i \\ 1+i & -\sqrt{2}i \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (0.64)$$

$$\left(\begin{array}{cc|c} -\sqrt{2}i & 1+i & 0 \\ 1+i & -\sqrt{2}i & 0 \end{array} \right) \xleftrightarrow{R_2 \rightarrow R_2 - \frac{1+i}{-\sqrt{2}i} R_1} \left(\begin{array}{cc|c} -\sqrt{2}i & 1+i & 0 \\ 0 & 0 & 0 \end{array} \right) \quad (0.65)$$

Hence $y = \frac{\sqrt{2}i}{1+i}x$ and $\mathbf{v}_2 = \begin{pmatrix} 1 \\ \frac{\sqrt{2}i}{1+i} \end{pmatrix}$.

$$\lambda_1 = 1 - \sqrt{2}i, \quad \mathbf{v}_1 = \begin{pmatrix} 1 \\ -\frac{\sqrt{2}i}{1+i} \end{pmatrix} \quad (0.66)$$

$$\lambda_2 = 1 + \sqrt{2}i, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ \frac{\sqrt{2}i}{1+i} \end{pmatrix} \quad (0.67)$$

The Eigen vectors are **complex**

Examples

(b) Let

$$\mathbf{A} = \begin{pmatrix} 1 & i & 0 \\ i & 1 & i \\ 0 & i & 1 \end{pmatrix} \quad (0.68)$$

Here, $\mathbf{A} = \mathbf{A}^\top$ but $\mathbf{A} \neq \mathbf{A}^*$, so $\mathbf{A}$ is complex symmetric but not Hermitian.

Finding Eigenvalues

$$|\mathbf{A} - \lambda \mathbf{I}| = 0 \quad (0.69)$$

$$\begin{vmatrix} 1 - \lambda & i & 0 \\ i & 1 - \lambda & i \\ 0 & i & 1 - \lambda \end{vmatrix} = 0 \quad (0.70)$$

Examples

$$\begin{vmatrix} 1-\lambda & i & 0 \\ i & 1-\lambda & i \\ 0 & i & 1-\lambda \end{vmatrix} \xleftrightarrow{R_2 \rightarrow R_2 - \frac{i}{1-\lambda} R_1} \begin{vmatrix} 1-\lambda & i & 0 \\ 0 & \frac{(1-\lambda)^2+1}{1-\lambda} & i \\ 0 & i & 1-\lambda \end{vmatrix} \quad (0.71)$$

$$\xleftrightarrow{R_3 \rightarrow R_3 - \frac{i(1-\lambda)}{(1-\lambda)^2+1} R_2} \begin{vmatrix} 1-\lambda & i & 0 \\ 0 & \frac{(1-\lambda)^2+1}{1-\lambda} & i \\ 0 & 0 & \frac{(1-\lambda)^3+2(1-\lambda)}{(1-\lambda)^2+1} \end{vmatrix} \quad (0.72)$$

$$(1-\lambda) \left(\frac{(1-\lambda)^2+1}{1-\lambda} \right) \left(\frac{(1-\lambda)^3+2(1-\lambda)}{(1-\lambda)^2+1} \right) = 0 \quad (0.73)$$

Simplifying,

$$(1-\lambda)((1-\lambda)^2+2) = 0 \quad (0.74)$$

Hence eigenvalues are:

$$\lambda_1 = 1, \lambda_2 = 1 + i\sqrt{2}, \lambda_3 = 1 - i\sqrt{2} \quad (0.75)$$

Examples

Finding Eigenvectors

For $\lambda_1 = 1$:

$$(\mathbf{A} - \mathbf{I})\mathbf{v} = 0 \quad (0.76)$$

$$\begin{pmatrix} 0 & i & 0 \\ i & 0 & i \\ 0 & i & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad (0.77)$$

$$\left(\begin{array}{ccc|c} 0 & i & 0 & 0 \\ i & 0 & i & 0 \\ 0 & i & 0 & 0 \end{array} \right) \xleftrightarrow{R_2 \leftrightarrow R_1} \left(\begin{array}{ccc|c} i & 0 & i & 0 \\ 0 & i & 0 & 0 \\ 0 & i & 0 & 0 \end{array} \right) \xleftrightarrow{R_3 \rightarrow R_3 - R_2} \left(\begin{array}{ccc|c} i & 0 & i & 0 \\ 0 & i & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \quad (0.78)$$

Hence $y = 0$, $z = -x \Rightarrow \mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$.

Examples

For $\lambda_2 = 1 + i\sqrt{2}$

$$\mathbf{A} - \lambda_2 \mathbf{I} = \begin{pmatrix} -i\sqrt{2} & i & 0 \\ i & -i\sqrt{2} & i \\ 0 & i & -i\sqrt{2} \end{pmatrix} \quad (0.79)$$

Forming the augmented matrix :

$$\left(\begin{array}{ccc|c} -i\sqrt{2} & i & 0 & 0 \\ i & -i\sqrt{2} & i & 0 \\ 0 & i & -i\sqrt{2} & 0 \end{array} \right) \quad (0.80)$$

$$\xleftrightarrow{R_2 \rightarrow R_2 - \frac{i}{-i\sqrt{2}} R_1} \left(\begin{array}{ccc|c} -i\sqrt{2} & i & 0 & 0 \\ 0 & i\left(\frac{1}{\sqrt{2}} - \sqrt{2}\right) & i & 0 \\ 0 & i & -i\sqrt{2} & 0 \end{array} \right) \quad (0.81)$$

$$\xleftrightarrow{R_3 \rightarrow R_3 - \frac{i}{i\left(\frac{1}{\sqrt{2}} - \sqrt{2}\right)} R_2} \left(\begin{array}{ccc|c} -i\sqrt{2} & i & 0 & 0 \\ 0 & i\left(\frac{1}{\sqrt{2}} - \sqrt{2}\right) & i & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \quad (0.82)$$

Examples

$$\mathbf{v}_2 = \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix} \quad (0.83)$$

For $\lambda_3 = 1 - i\sqrt{2}$

$$\mathbf{A} - \lambda_3 \mathbf{I} = \begin{pmatrix} i\sqrt{2} & i & 0 \\ i & i\sqrt{2} & i \\ 0 & i & i\sqrt{2} \end{pmatrix} \quad (0.84)$$

Forming the augmented matrix:

$$\left(\begin{array}{ccc|c} i\sqrt{2} & i & 0 & 0 \\ i & i\sqrt{2} & i & 0 \\ 0 & i & i\sqrt{2} & 0 \end{array} \right) \xleftrightarrow{R_2 \rightarrow R_2 - \frac{i}{i\sqrt{2}} R_1} \left(\begin{array}{ccc|c} i\sqrt{2} & i & 0 & 0 \\ 0 & i \left(\sqrt{2} - \frac{1}{\sqrt{2}} \right) & i & 0 \\ 0 & i & i\sqrt{2} & 0 \end{array} \right) \quad (0.85)$$

Examples

$$\begin{array}{c} R_3 \rightarrow R_3 - \frac{i}{i\left(\sqrt{2} - \frac{1}{\sqrt{2}}\right)} R_2 \\ \longleftrightarrow \end{array} \left(\begin{array}{ccc|c} i\sqrt{2} & i & 0 & 0 \\ 0 & i\left(\sqrt{2} - \frac{1}{\sqrt{2}}\right) & i & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \quad (0.86)$$

Hence, $y = -\sqrt{2}x$, $x = z$.

$$\mathbf{v}_3 = \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix} \quad (0.87)$$

Examples

Thus, by Gaussian elimination:

$$\lambda_1 = 1, \quad \mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \quad (0.88)$$

$$\lambda_2 = 1 + i\sqrt{2}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix} \quad (0.89)$$

$$\lambda_3 = 1 - i\sqrt{2}, \quad \mathbf{v}_3 = \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix} \quad (0.90)$$

Some of the eigen values are complex but the eigen vectors are **real**.

Examples

(c)

$$\mathbf{A} = \begin{pmatrix} 1 & i \\ 0 & 2 \end{pmatrix} \quad (0.91)$$

$$|\mathbf{A} - \lambda \mathbf{I}| = \begin{vmatrix} 1 - \lambda & i \\ 0 & 2 - \lambda \end{vmatrix} = (1 - \lambda)(2 - \lambda) \quad (0.92)$$

$$\lambda_1 = 1, \quad \lambda_2 = 2 \quad (0.93)$$

For $\lambda_1 = 1$,

$$\mathbf{A} - \lambda_1 \mathbf{I} = \begin{pmatrix} 0 & i \\ 0 & 1 \end{pmatrix} \quad (0.94)$$

$$iy = 0 \Rightarrow y = 0 \quad (0.95)$$

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (0.96)$$

Examples

For $\lambda_2 = 2$,

$$\mathbf{A} - \lambda_2 \mathbf{I} = \begin{pmatrix} -1 & i \\ 0 & 0 \end{pmatrix} \xrightarrow{R_1/(-1)} \begin{pmatrix} 1 & -i \\ 0 & 0 \end{pmatrix} \quad (0.97)$$

$$x = iy \quad (0.98)$$

$$\mathbf{v}_2 = \begin{pmatrix} i \\ 1 \end{pmatrix} \quad (0.99)$$

Check orthogonality:

$$\mathbf{v}_1^* \mathbf{v}_2 \neq 0 \quad (0.100)$$

Hence, the eigenvectors are not orthogonal.